

1.7 Excavation & Backfill Log

The permanent record of what came out of the hole and what went back in — depths, soil conditions, water, compaction, and final grade.

Project Name: _____ Excavation Contractor: _____

Property Address: _____ Contact Phone: _____

Owner-Builder: _____ Equipment Used: _____

PURPOSE: This log documents all excavation and backfill activities for the foundation. Accurate records are essential for verifying code compliance, documenting soil conditions, and supporting warranty claims. Take photos at each stage.

SECTION 1: PRE-EXCAVATION INFORMATION

Pre-Excavation Checklist

	Pre-Excavation Checklist	Date Completed	Notes
☐	811 utility locate completed Ticket #: _____		
☐	All utility locations marked and photographed		
☐	Building permit approved and posted on site		
☐	Foundation layout staked and squared		
☐	Erosion control measures in place		
☐	Tree protection fencing installed (if applicable)		

SECTION 2: EXCAVATION WORK LOG

Excavation Schedule

Excavation start date: _____ Start time: _____

Excavation completion date: _____ Completion time: _____

Total excavation duration: _____ Weather conditions: _____

Excavation Depth Measurements

Required excavation depth per plan (inches below finished grade): _____

Local frost depth requirement (inches): _____

Excavation Depth Verification

Measurement Location	Required Depth (in)	Actual Depth (in)	Pass / Fail	Notes
Northeast (NE) Corner			☐ Pass ☐ Fail	
Northwest (NW) Corner			☐ Pass ☐ Fail	
Southeast (SE) Corner			☐ Pass ☐ Fail	
Southwest (SW) Corner			☐ Pass ☐ Fail	
North Wall Midpoint			☐ Pass ☐ Fail	
South Wall Midpoint			☐ Pass ☐ Fail	
East Wall Midpoint			☐ Pass ☐ Fail	
West Wall Midpoint			☐ Pass ☐ Fail	
Center of Excavation			☐ Pass ☐ Fail	
Other: _____			☐ Pass ☐ Fail	

SECTION 3: SOIL CONDITIONS DOCUMENTATION

Soil Type & Classification

Primary soil type: ☐ Clay ☐ Silt ☐ Sand ☐ Gravel ☐ Loam ☐ Mixed

Other (describe): _____

Soil color: _____

Soil moisture level:

☐ Dry ☐ Slightly moist ☐ Wet ☐ Saturated ☐ Standing water

Soil bearing capacity (PSF, per geotechnical report or code assumption): _____

Organic matter present:

☐ None ☐ Minimal ☐ Moderate ☐ Significant (requires removal)

Frost-susceptible soil: ☐ Yes ☐ No

Action taken: _____

Rock, Ledge & Obstacles Encountered

Location	Depth Found (in)	Type (rock, ledge, debris)	Action Taken	Notes

Water & Drainage Issues

Standing water encountered: ☐ Yes ☐ No

If yes — location: _____

Groundwater seepage: ☐ Yes ☐ No

If yes — depth (inches): _____

Flow rate: ☐ Slow ☐ Moderate ☐ Fast

Dewatering required: ☐ Yes ☐ No

Method: ☐ Pump ☐ Sump ☐ Gravity drain

Other method: _____

High water table noted: ☐ Yes ☐ No

Depth from surface (feet): _____

Drainage Improvements Made

Excavated Material Disposition

Material Type	Est. Volume (cu. yds)	Disposition	Location / Notes
Topsoil (saved for landscaping)		☐ Stockpiled ☐ Removed	
Subsoil (reusable for backfill)		☐ Stockpiled ☐ Removed	
Rock/gravel		☐ Stockpiled ☐ Removed	
Unsuitable material (organic, wet)		☐ Removed ☐ Hauled away	

SECTION 4: BACKFILL OPERATIONS LOG

Backfill Schedule

Backfill start date: _____ Days since concrete pour: _____

Backfill completion date: _____ Weather conditions: _____

Waterproofing cured: ☐ Yes ☐ No ☐ N/A

Drainage tile installed: ☐ Yes ☐ No ☐ N/A

Backfill Material Specifications

Primary backfill material:

☐ Excavated soil ☐ Clean fill ☐ Sand ☐ Gravel ☐ Crushed stone

Other: _____

Material source: ☐ On-site ☐ Imported

Supplier (if imported): _____

Material quality (check all that apply):

☐ Free of organic matter ☐ Free of large rocks (>3") ☐ Properly graded

Moisture content: ☐ Optimum for compaction ☐ Too wet ☐ Too dry

Action taken (if too wet or too dry): _____

Total backfill volume (cubic yards): _____

Compaction Method & Testing

Compaction equipment used:

☐ Plate compactor ☐ Jumping jack ☐ Roller ☐ Hand tamper

Other equipment: _____

Lift thickness (in): _____ Passes per lift: _____

Total number of lifts: _____ Target compaction (%): _____

Typical values: lifts of 6-12 inches before compaction; target compaction 90-95% of maximum dry density for backfill.

Compaction testing performed: ☐ Yes ☐ No

Test method: ☐ Nuclear density ☐ Proctor ☐ Visual

Other method: _____

Compaction Test Results

Test Location	Lift Depth (in)	Compaction % Achieved	Pass / Fail	Notes / Actions
NE Corner			☐ Pass ☐ Fail	
NW Corner			☐ Pass ☐ Fail	

Compaction Test Results

Test Location	Lift Depth (in)	Compaction % Achieved	Pass / Fail	Notes / Actions
SE Corner			☐ Pass ☐ Fail	
SW Corner			☐ Pass ☐ Fail	
North Wall			☐ Pass ☐ Fail	
South Wall			☐ Pass ☐ Fail	
East Wall			☐ Pass ☐ Fail	
West Wall			☐ Pass ☐ Fail	

SECTION 5: FINAL GRADING & DRAINAGE

Final Grading Checklist

	Final Grading Checklist	Date Completed	Notes
☐	Backfill completed to within 6-8" of final grade		
☐	Grade slopes away from foundation (2% min, 6" drop in 10 feet)		
☐	No low spots or depressions where water can collect		
☐	Swales or drainage paths direct water away from structure		
☐	Perimeter drain outlet is clear and functional		
☐	Final topsoil layer applied (if applicable)		
☐	Erosion control measures in place until landscaping		
☐	Grading verified after first significant rain event		

SECTION 6: PHOTO DOCUMENTATION

Photographic evidence is critical for insurance claims, warranty issues, and inspection appeals. Take multiple photos from different angles at each stage.

Photo Documentation Log

Date	Photo Description	File Name / Number	Notes
	Site before excavation begins		
	Utility locate markings		
	Foundation layout stakes/strings		
	Excavation in progress		
	Final excavation depth verification		
	Soil conditions/layers exposed		
	Rock or unusual conditions found		
	Water/drainage issues encountered		
	Foundation complete before backfill		
	Waterproofing applied		
	Drainage tile installation		
	Backfill in progress (show lifts)		
	Compaction equipment in use		
	Final backfill completed		
	Final grading and drainage		

SECTION 7: CONTRACTOR SIGN-OFF & CERTIFICATION

I certify that all excavation and backfill work has been completed in accordance with approved plans and industry best practices. All compaction meets or exceeds specified requirements.

Excavation Contractor

Signature: _____ Date: _____

Name (printed): _____ Company: _____

License # (if applicable): _____

Owner-Builder Acceptance

Signature: _____ Date: _____

NOTES, ISSUES & LESSONS LEARNED
